

19  $D = \frac{u^2}{2a}$

$u = 26.2$  correct to 3 significant figures

$a = 4.3$  correct to 2 significant figures

- (a) Calculate the upper bound for the value of  $D$ .  
Give your answer correct to 6 significant figures.  
You must show all your working.

(3)

The lower bound for the value of  $D$  is 78.6003 correct to 6 significant figures.

- (b) By considering bounds, write down the value of  $D$  to a suitable degree of accuracy.  
You must give a reason for your answer.

(2)